

NETS IN SEPARABLE $\mathcal{L}_\infty$ SPACES

RUN FA_BANACH_001

Status. `literature_implied_answer` (partial subcase). This is not a new proof and not a full solution of Kalton’s question. It records that a later theorem of Hajek–Medina gives a positive Schauder-basis answer for nets in separable infinite-dimensional $\mathcal{L}_\infty$ -spaces.

SOURCE QUESTION

Hajek–Medina, *Schauder bases in Lipschitz free spaces over nets in Banach spaces*, arXiv:2112.03095, recall Kalton’s question whether Lipschitz free spaces over separable uniformly discrete metric spaces have the bounded approximation property. They note that it suffices to consider the case where the metric space is a net N in a Banach space X , that Kalton proved a positive answer when X has separable dual or is a separable dual space, and that the case of a general separable Banach space X remains open.

The same source paper proves stronger Schauder-basis conclusions in several special cases, including finite-dimensional spaces and Banach spaces with a Schauder basis containing c_0 , or with a c_0 -like FDD.

SUPPORTING LITERATURE

Hajek–Medina, *Schauder bases in Lipschitz free spaces over nets of $\mathcal{L}_\infty$ -spaces*, arXiv:2202.08032, explicitly says that it is motivated by the same Kalton question and studies the retractional structure of nets in $\mathcal{L}_\infty$ -spaces. Its introduction states that, as a corollary of the construction, the Lipschitz free space $\mathcal{F}(N)$ over a net in any separable $\mathcal{L}_\infty$ -space has a Schauder basis.

The decisive statements are the paper’s *Main Theorem* and the immediate corollary:

a net in any separable infinite-dimensional $\mathcal{L}_\infty$ -space has a retractional basis; hence, if N is such a net, then $\mathcal{F}(N)$ has a Schauder basis.

The first implication from a retractional basis to a Schauder basis is also recorded in the same paper via the cited retractional-basis theorem.

Literature-implied subcase. *Let X be a separable infinite-dimensional $\mathcal{L}_\infty$ -space and let N be a net in X . Then $\mathcal{F}(N)$ has a Schauder basis. In particular, $\mathcal{F}(N)$ has the bounded approximation property.*

Proof. The theorem and corollary in arXiv:2202.08032 give the Schauder basis directly. The bounded approximation property follows because the partial-sum projections associated to a Schauder basis are finite-rank operators and are uniformly bounded by the basis constant. $\square$

SCOPE

This is a proper subcase of the source target, not a solution of the general separable Banach-space case. The packet records a positive answer for nets in separable infinite-dimensional $\mathcal{L}_\infty$ -spaces. Finite-dimensional and c_0/c_0 -like FDD cases are already handled by the source paper arXiv:2112.03095 itself.

SEARCH AND PROVENANCE NOTES

The run indexes were searched for arXiv:2112.03095, the source title, free spaces over nets, Schauder bases, retractional bases, and the BAP question. No prior packet or attempt for this exact target or this $\mathcal{L}_\infty$ -subcase was found. External search found the decisive later paper arXiv:2202.08032, along with adjacent but nondecisive papers on retractional-basis obstructions and Lipschitz-free spaces over nets.

The supporting paper is by the same authors and explicitly says it is motivated by the Kalton/net BAP problem and extends their previous net-basis work. This packet classifies the relation as an agent-identified literature-implied partial subcase because the general separable Banach-space problem remains open.

REFERENCES

- [1] P. Hajek and R. Medina, *Schauder bases in Lipschitz free spaces over nets in Banach spaces*, arXiv:2112.03095.
- [2] P. Hajek and R. Medina, *Schauder bases in Lipschitz free spaces over nets of $\mathcal{L}_\infty$ -spaces*, arXiv:2202.08032.